LANDSCAPE ARCHITECTURE PORTFOLIO

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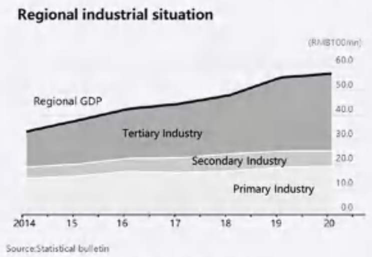
LIGHT THE LAMP OF FIREFLIES ——BAOTING ECOLOGICAL AND INDUSTRIAL TRANSFORMATION

BACKGROUND ANALYSIS

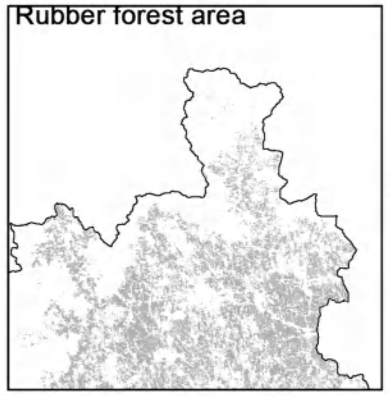
1.INDUSTRIAL STRUCTURE ANALYSIS



- •GDP growth is stable
- •The tertiary industry accounts for a large proportion and rises rapidly
- •The proportion of tertiary industry is very low
- •The primary industry increases slightly

AGRICULTURE

58% Rubber	Area 18900ha	Yield 13kt
16% Areca	Area 5350ha	Yield 15kt
14% Fruits	Area 4650ha	Yield 24kt
12% Gain	Area 3860ha	Yield 14kt



- •Large area rubber forest
- •Extensive management
- •1500 years history of areca planting

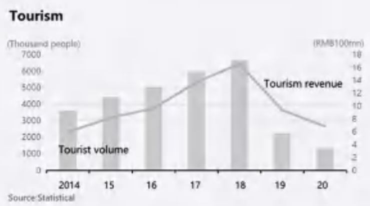
INDUSTRY

84% Construction	94M CNY	
16% Industry	507M CNY	Added value

3904t COD → Clean
457t NO_x, 2t SO_2 → Fresh
Clean, Fresh → Habitat
Expending × Habitat

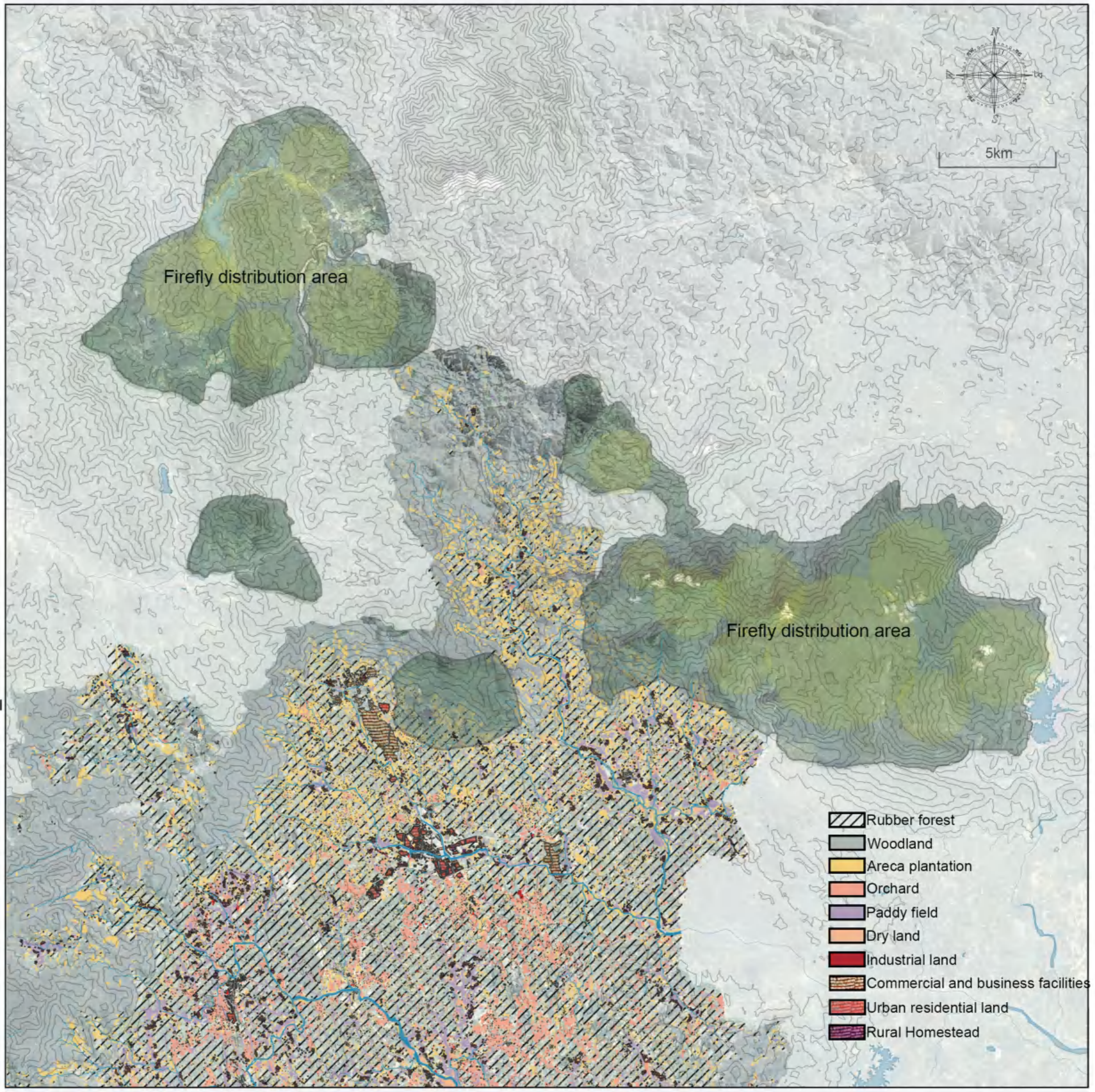
- •Low proportion of industry
- •Has an environmental basis for firefly habitat
- •Habitat is losing

TOURISM

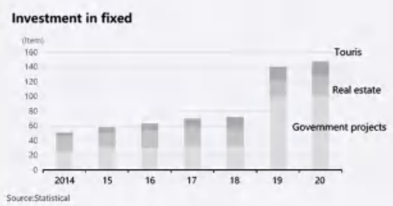


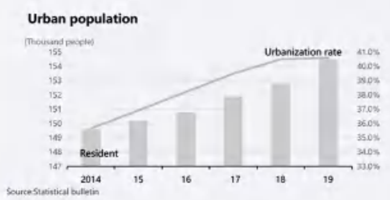
Rainforest, Hot spring, Regimen → Ecotourism

- •Environment friendly
- •Great development prospects
- •Government support
- •Affected by the COVID-19



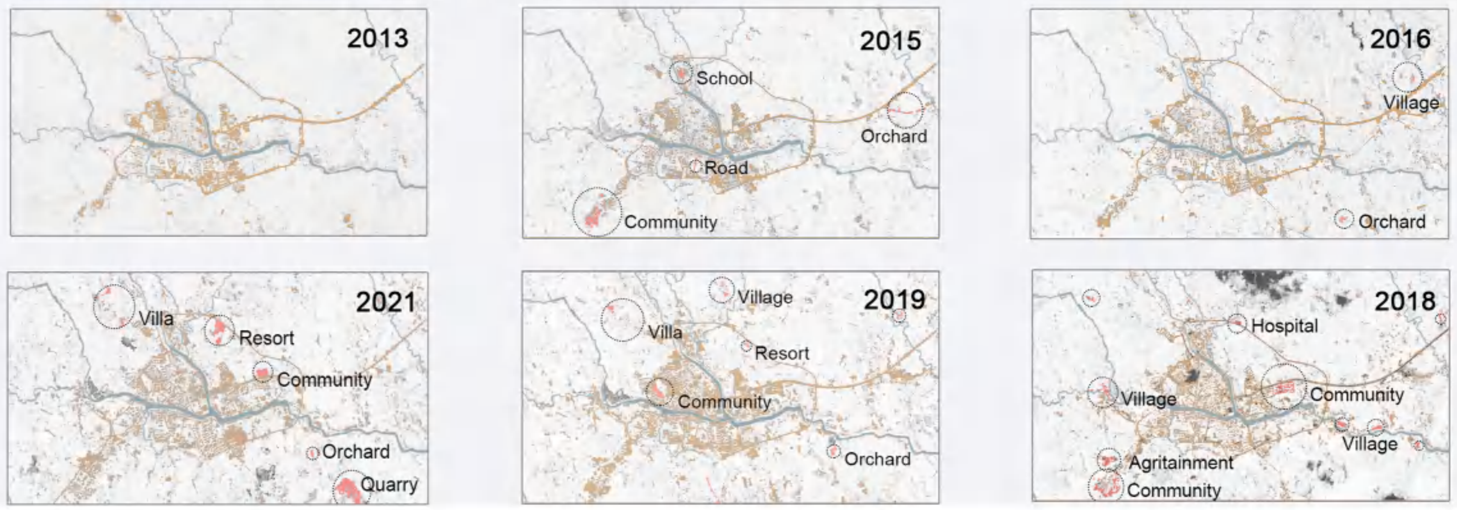
2.ANALYSIS OF RESIDENCE



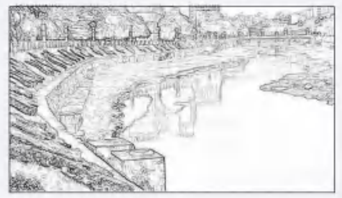


- •Tourism investment projects have increased year by year
- •Real estate, at around 60% of total investment, simply takes up too much economic room
- •Government investment projects increased after the COVID-19
- •The improvement of population and urbanization has brought about the demand for housing and employment

3.HISTORICAL EXPANSION OF TOWN



4.EXISTING CONDITIONS


Clean river

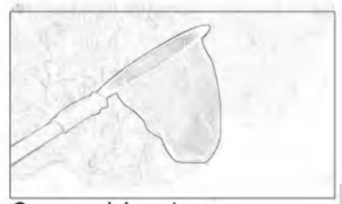

Clear atmosphere


Ecotourism strategy


Herbicide application


Soil degradation

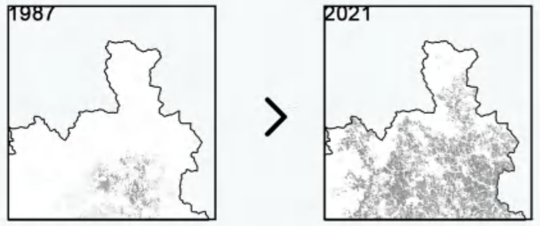

Town lights


Commercial capture

PRESENT SITUATION ANALYSIS

RUBBER FOREST EXPANSION

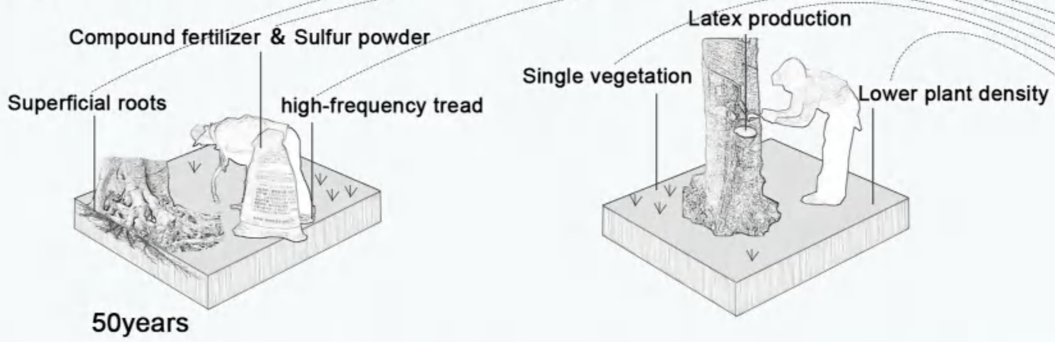
1.CHANGE



1987~2021 area of Rubber monoculture

x 6

2.PROBLEM



3.INDICATOR

- Water holding capacities
- pH
- Total porosity
- Soil nutrients
- Water content
- Soil bulk density

4.CONSEQUENCE

- Biodiversity and biomass loss
- Excessive soil erosion and acidification
- Decline in soil fertility
- Carbon sequestration
- Landslide

TROPICAL RAIN FOREST CRISIS

100years | 50years | 20years

1.TROPICAL RAIN FOREST

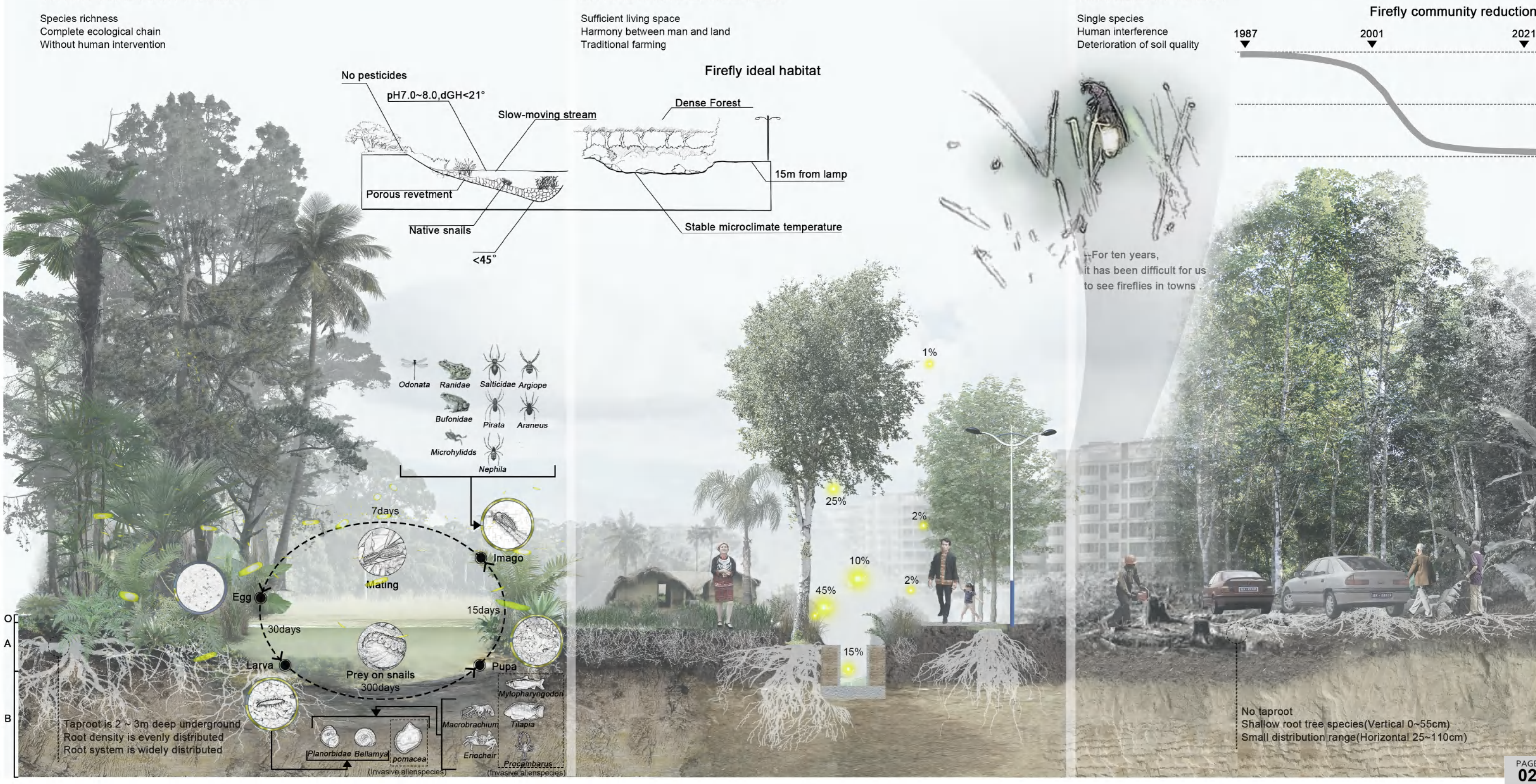
Species richness
Complete ecological chain
Without human intervention

2.TRADITIONAL VILLAGES

Sufficient living space
Harmony between man and land
Traditional farming

3.RUBBER FOREST

Single species
Human interference
Deterioration of soil quality



STRATEGY
COMPREHENSIVE IMPROVEMENT PLAN

Module	Target	Problem
Paddy field, Economic forest	Transitional area	Pesticide and herbicide abuse; Single tree species in economic forests; Lack of boundary constraints
Residential area	Isolation of human impacts	Light pollution; Real estate development expansion
Activity area	Firefly ecotourism	Commercial capture of fireflies; Disturbance from tourists
Conservation area	Tropical Rainforest Habitat	Habitat encroachment; Lack of effective planning and management

Expect

Step1 Basic governance

General planning

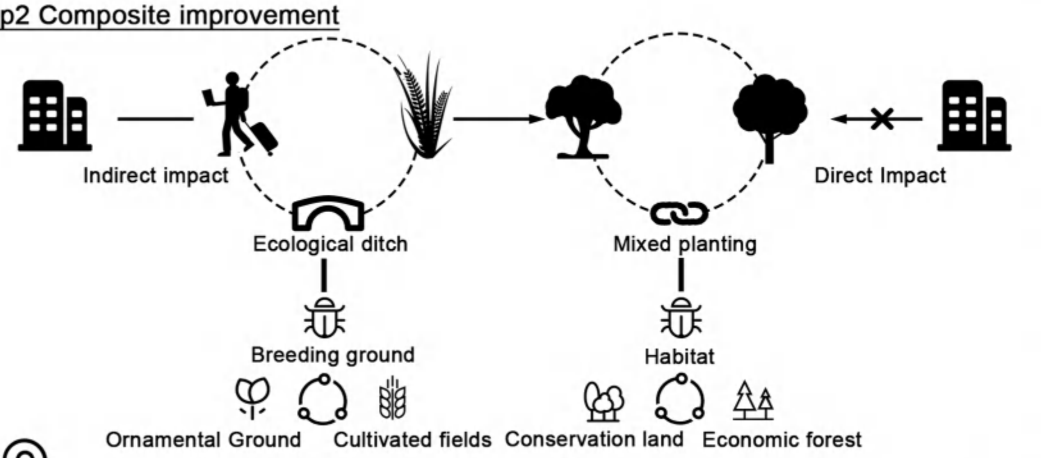
①Natural reserve
Monitoring and managing fireflies

②Transitional area
Developing organic agriculture

③Tourist area
Tourism infrastructure construction

④Urban area
Planning expansion scope

Step2 Composite improvement

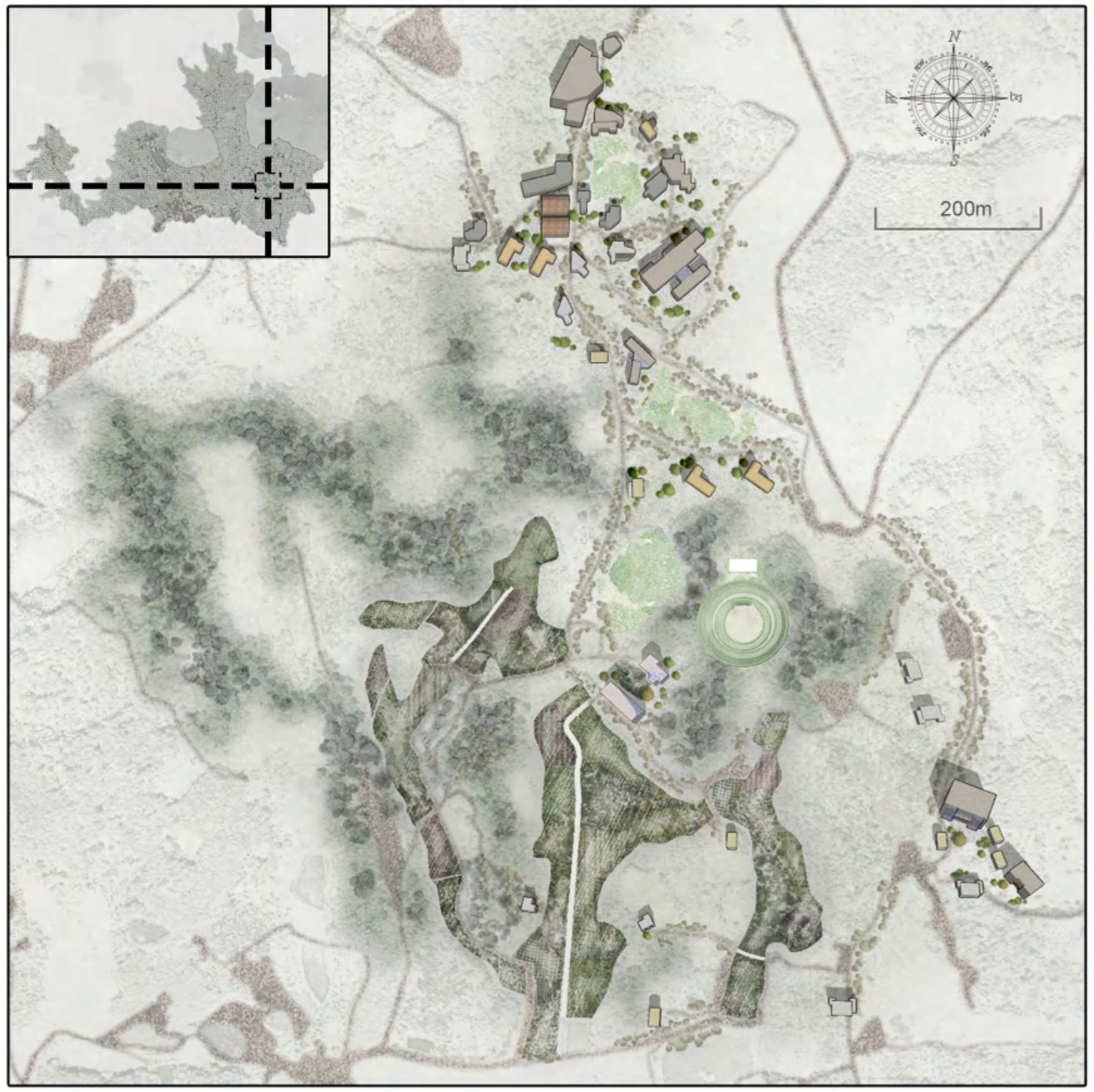


Key point

- Clean water with slow flow rate
- Space for reproduction and incubation
- Insect protection facilities
- Suitable viewing methods
- Mixed planting of multiple local plants
- Enclosed protection zone
- Establishing complex tree roots
- Preventing excessive manual intervention

Zoning plan of Pailiao Village

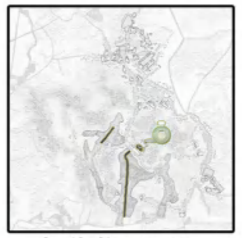
Households382 Population1543 Economic forest351.3ha Paddy field51.3ha




Forest area

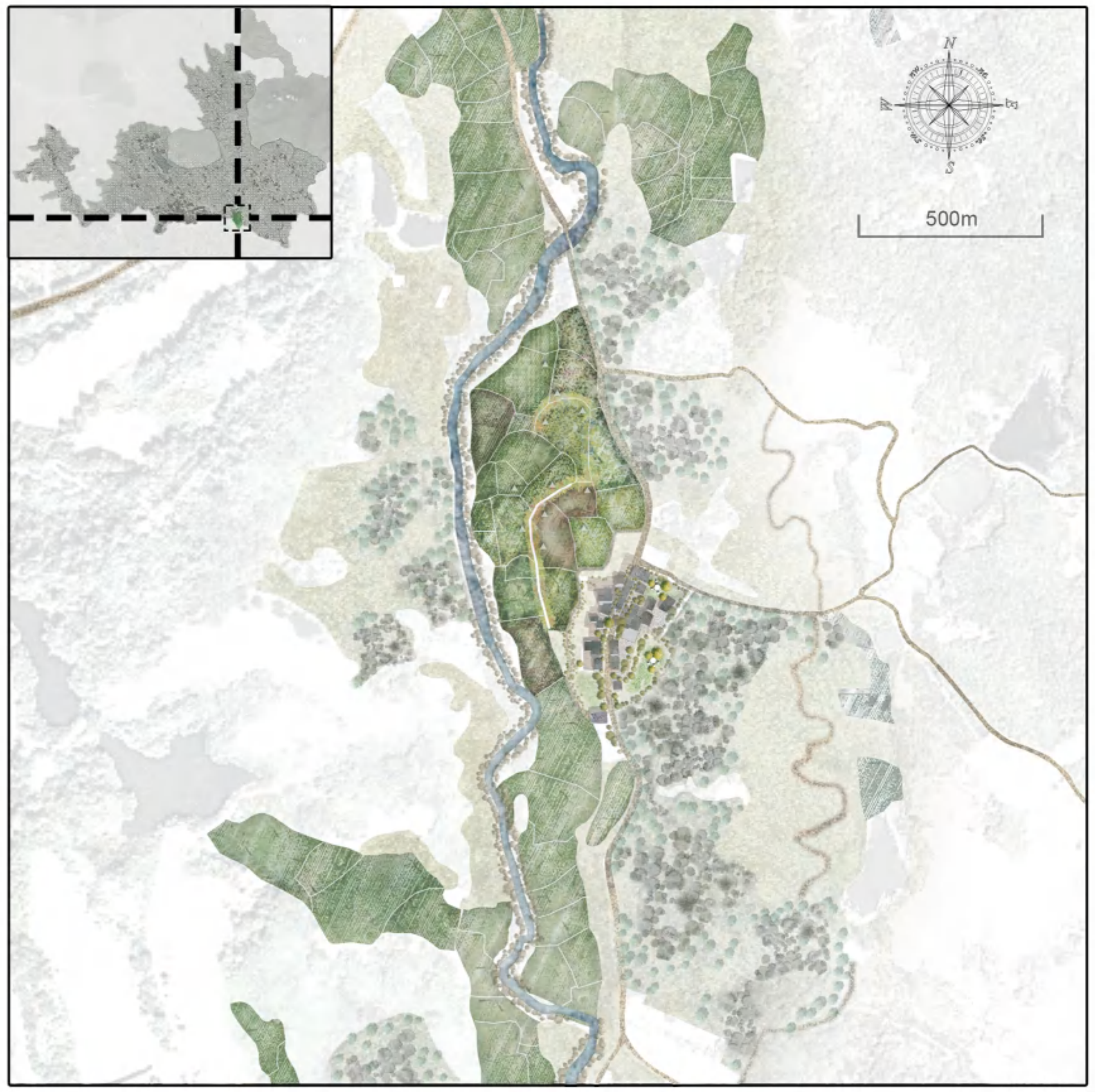

Paddy field


Residential area


Activity area

Zoning plan of Shidong Village

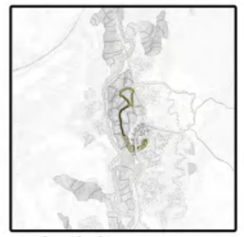
Households378 Population1568 Economic forest287.4ha Paddy field72.3ha




Forest area


Paddy field

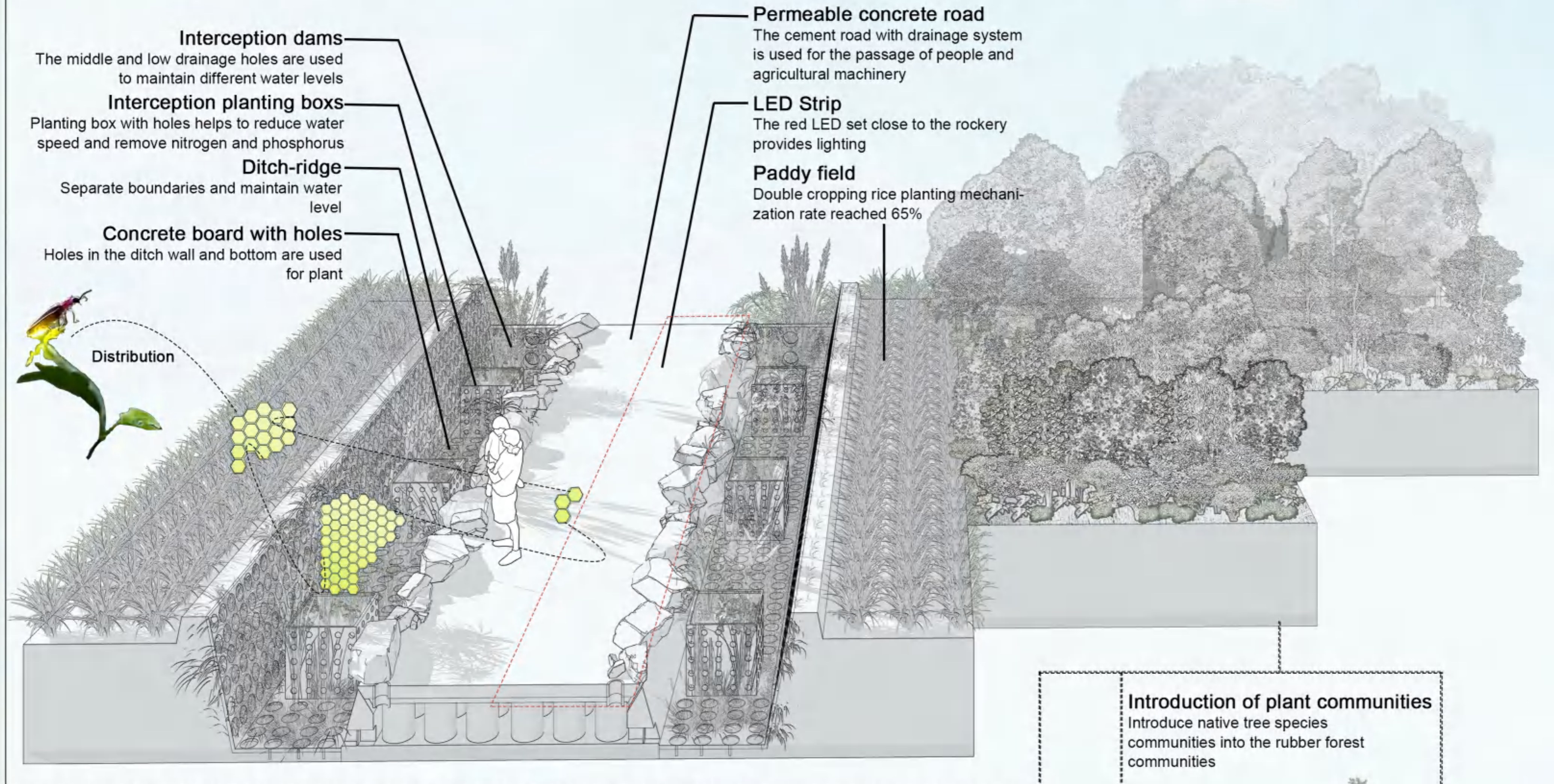

Residential area

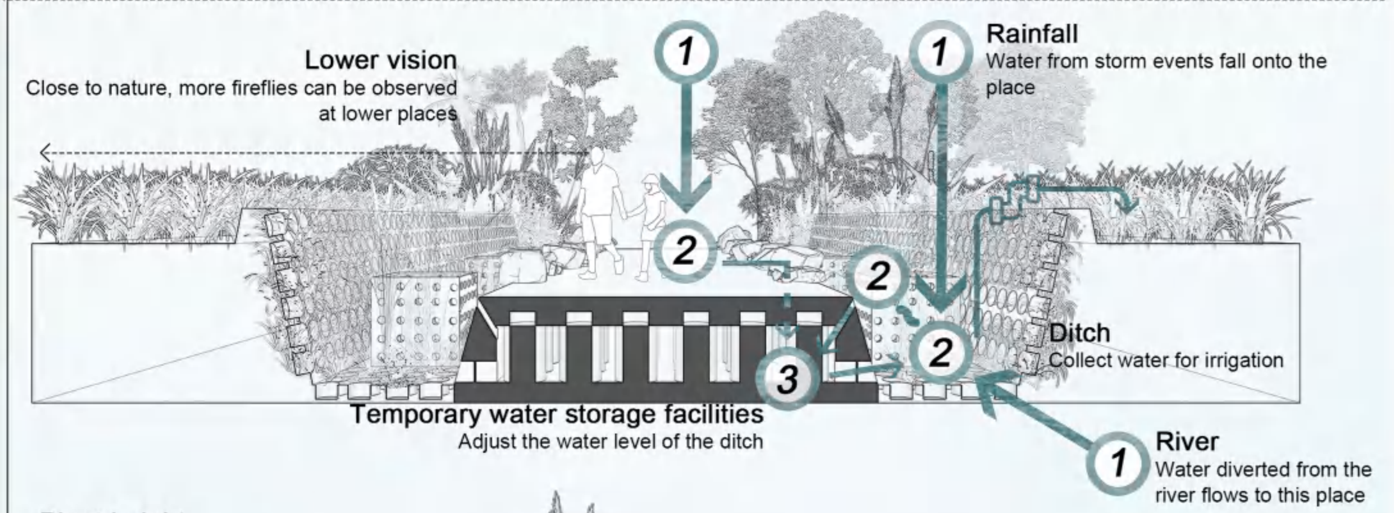

Activity area

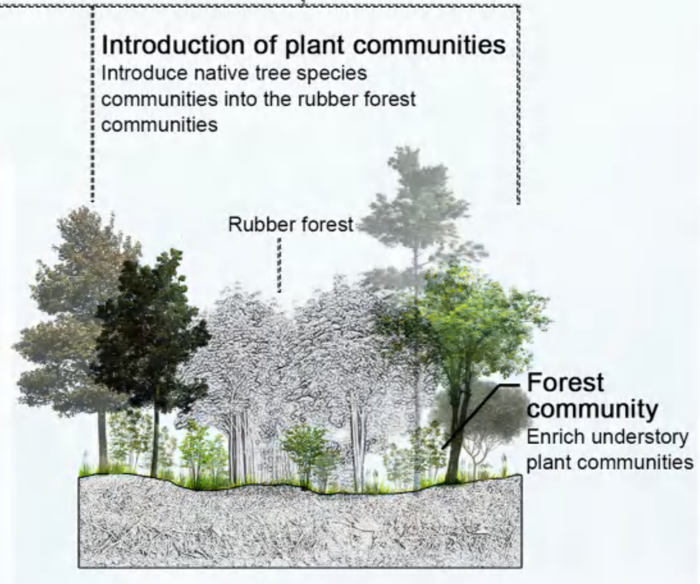
SCHEME DESIGN

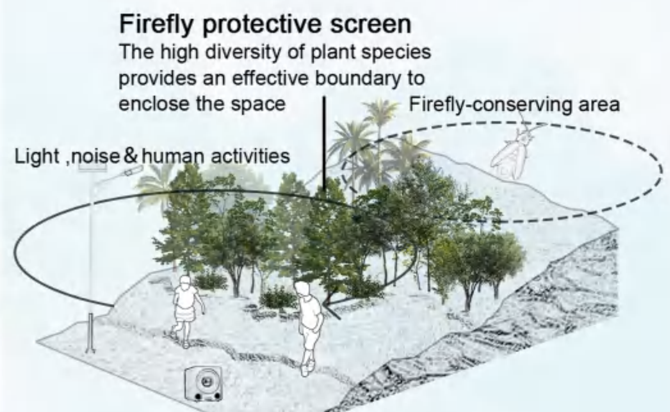
1.FIREFLY-SUITABLE DITCH SYSTEM

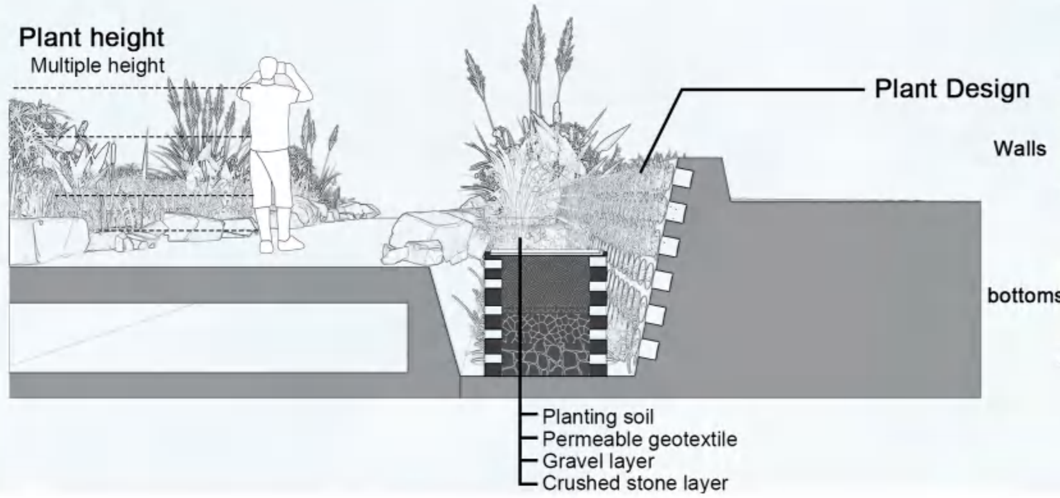
2.DIVERSIFICATION OF PLANT COMMUNITIES

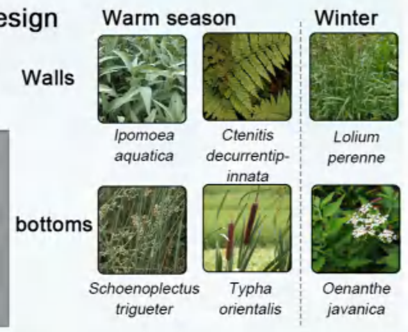






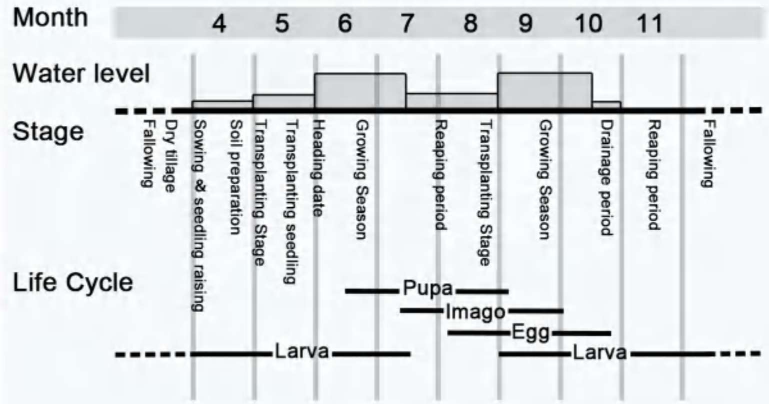






3.FIREFLIES' USE OF DESIGN

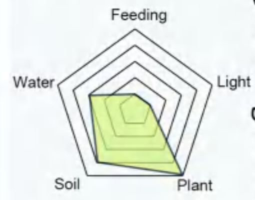
SCHEDULE OF DOUBLE CROPPING OF RICE AND FIREFLIES



ADAPT TO THE LIFE CYCLE OF FIREFLIES

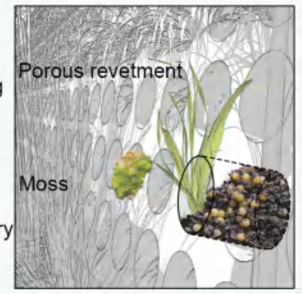
Environmental requirements | **Utilization mode**

Egg

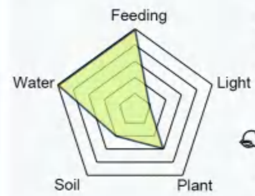


Soil
Prevent fungi causing egg death in soil

Plant
Moss, water grass and fallen leaves are necessary for spawning

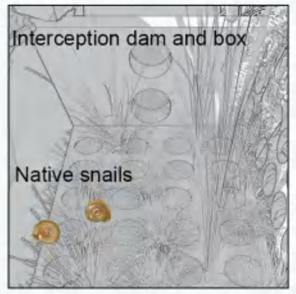


Larva

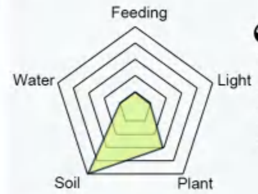


Water
Slow water flow, and keep water clear without excessive nitrogen and phosphorus nutrition

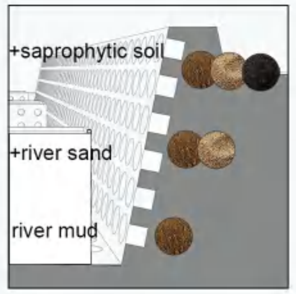
Feeding
Provide conditions to maintain sufficient local snails as a source of food, and prevent exotic snails,shrimp or fish



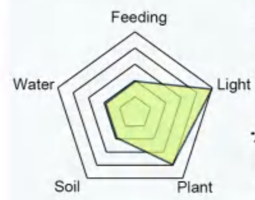
Pupa



Soil
Porous revetment and composite natural soil provide pupation environmental conditions

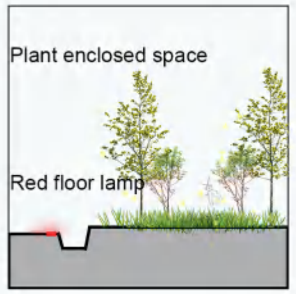


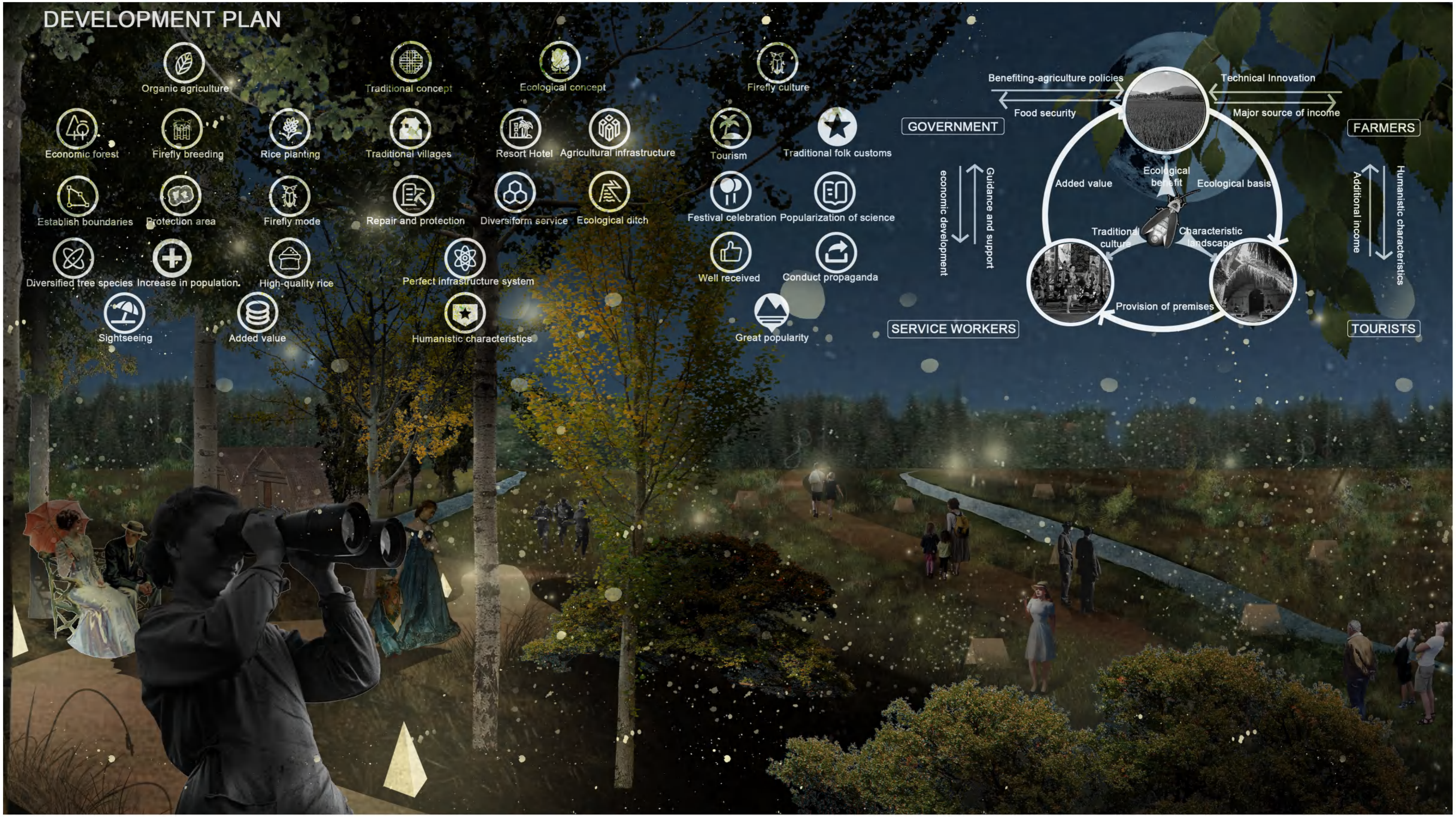
Imago



Plant
Safe breeding place enclosed by trees, shrubs and grasses

Light
Limit the light range and turn it into red light to reduce the interference of artificial light on courtship





1 FIREFLY-SUITABLE AGRICULTURE

2 FIREFLY LANDSCAPE STREET

3 FIREFLY FEATURED TOURISM

HOMELAND BUILT IN HURRICANES
——ESTABLISHMENT OF WIND AND FLOOD RESISTANT COMMUNITIES IN MOZAMBIQUE

BACKGROUND ANALYSIS

Recurrent climate-related shocks such as hurricanes and the floods they bring continue to have detrimental effects on the lives, property, and food security of communities across Mozambique, significantly increasing their vulnerability.

1 LOCATION

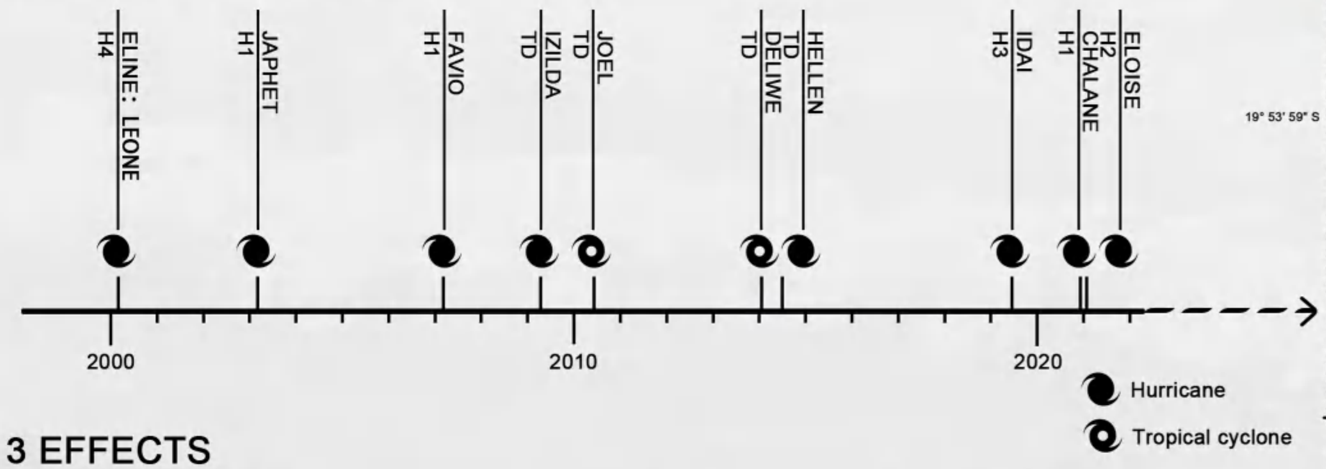
Location:

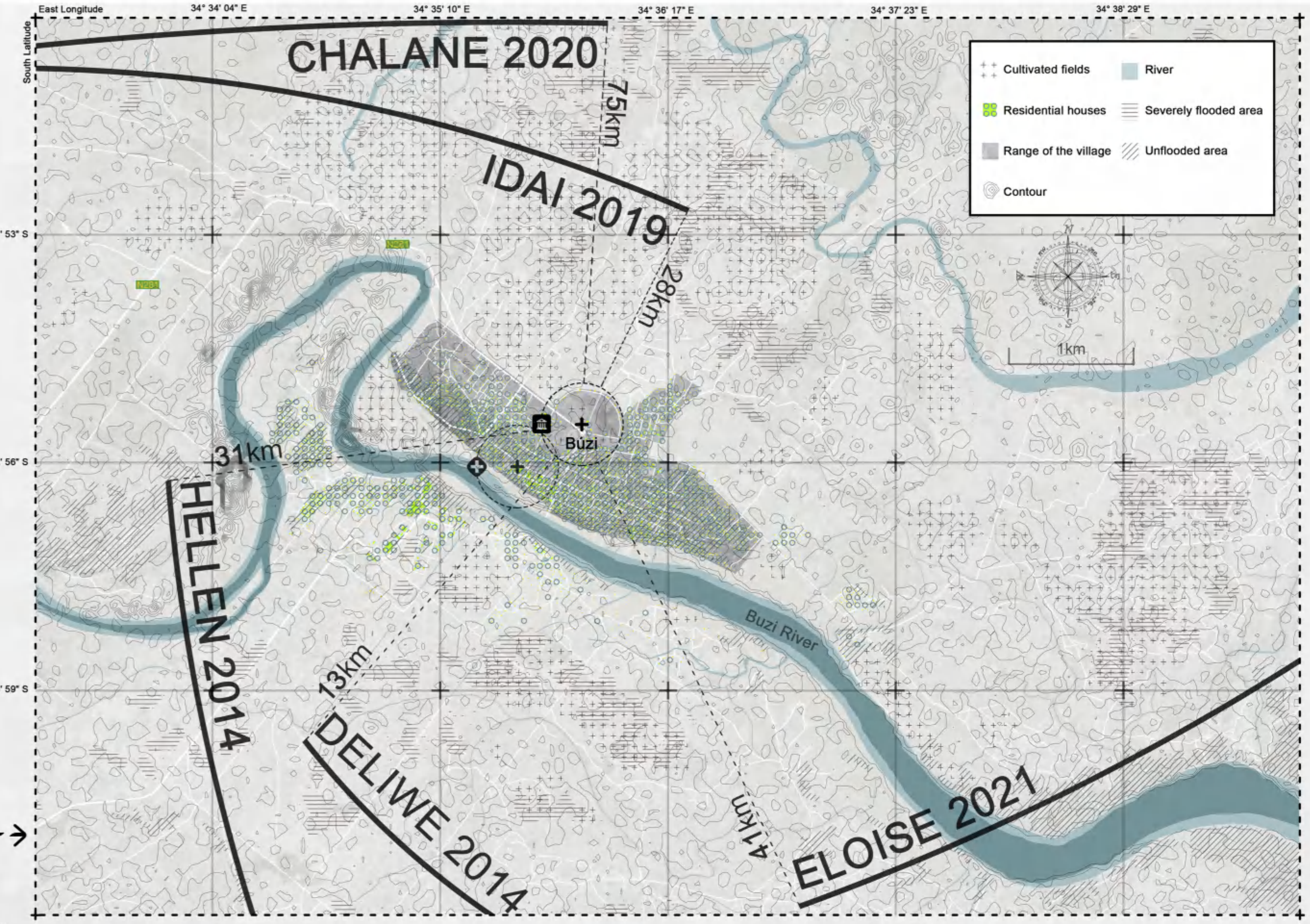
Mozambique

Buzi district

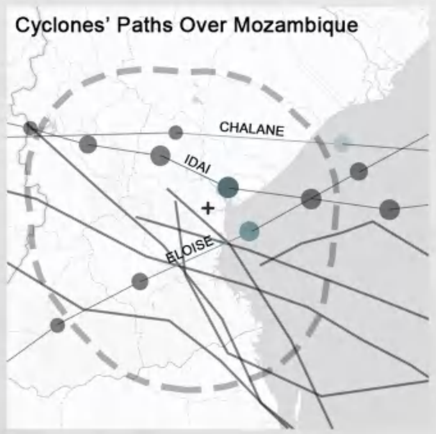
Villa Arriage

2 HURRICANE TIMELINE





3 EFFECTS



ELOISE
01/23/2021 DATE
90kt WIND SPEED
H2 CATEGORY
968mb PRESSURE

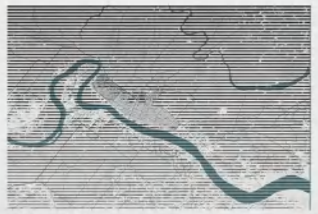
CHALANE
12/30/2020 DATE
70kt WIND SPEED
H1 CATEGORY
985mb PRESSURE

IDAI
03/15/2019 DATE
100kt WIND SPEED
H3 CATEGORY
965mb PRESSURE

4 INUNDATED AREA

IDAI

CHALANE



ELOISE

MOZAMBIQUE IN IDAI

603 DEATHS
400000 DISPLACED
1.8M IMPACTED
700000ha CROPS LAND IMPACTED
$773M DAMAGES
$258M IN CROPS DESTROYED
2165km² IN UNDATED AREA

BUZI IN IDAI

110 DEATHS
4060 TOTALLY
7420 PARTIALLY
73765ha CROPS LAND IMPACTED
7850 LOST CATTLE
25169 LOST POULTRY

5 DIFFICULTIES

Flooded land

Settled land

Transport infrastructure

Drinking water

Destroyed houses

Shelter

Daily necessities

Furniture

Food shortage

Food

Planting field

Pestilence

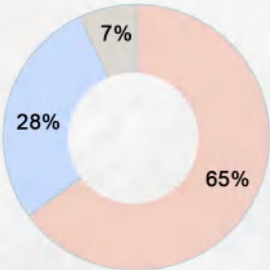
Medical supplies

Medical personnel

PRESENT SITUATION ANALYSIS

1 LOCAL-STRUCTURE DWELLING HOUSES

THE PROPORTION OF DIFFERENT STRUCTURES-BUILDINGS



STRUCTURAL MATERIAL ANALYSIS

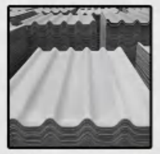
BRICK MASONRY HOUSE STRUCTURE | LIGHT STEEL HOUSE | ADOBE HOUSE

Brick masonry houses are supported by brick walls or columns

Light steel houses generally use internal transverse walls as load-bearing walls for the structure

The roof truss, roof, and lateral loads of the house are borne by the raw soil walls

MATERIAL


Asbestos tile


Corrugated steel


Thatch


Stone brick


Concrete floor


Wood strip


Wood


Angle iron


Immature soil

The bottom load-bearing component is a reinforced concrete flexible frame, while the upper load-bearing component is a rigid brick wall

The corner columns are welded by two C-shaped steel beams in pairs, located on the four corners of the house, and are hinged to the ground beam during installation; The middle column is made of two C-shaped steel bars welded back to back with long edges

Rammed earth wall is formed by placing semi wet and semi dry cohesive soil between wooden plywood and compacting it layer by layer and in sections

2 LOCAL AGRICULTURAL PRODUCTION

CROP TYPES

GRAIN CROPS


Maize


Cassava


Rise


Soybean

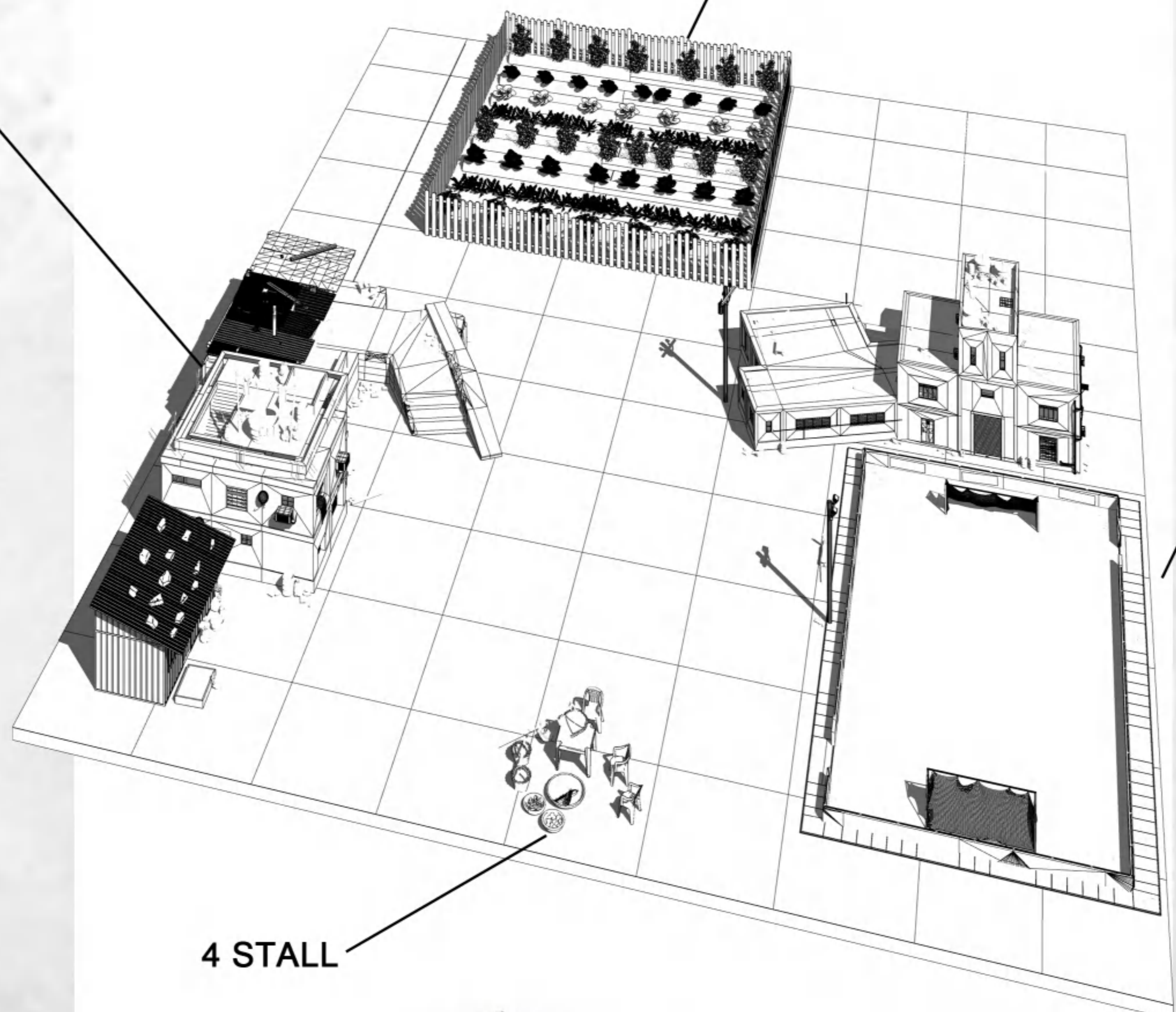
CASH CROPS


Cotton


Tobacco


Sesame

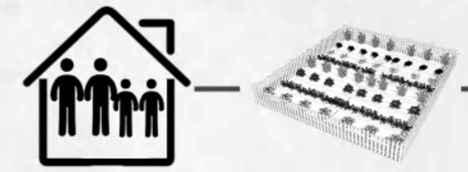

Sugarcane



4 STALL

Villagers trade agricultural products on the roadside, with small stalls on a household basis

SUBSISTENCE FARMING



Smallholder “family” sub-sector

- 97 % farmers without legal land title
- Backward agricultural technology
- Frequent floods and droughts

1

- Poor seed quality
- Poor seed-ling raising
- Low agricul-tural inputs

2

- No fertilizer
- Low technology
- Low mech-anization

3

- No irrigation systems
- Rainfed
- No farm chemical

4

- severe weather
- Transportation difficulties
- Low storage technology


Low grain yield


Poor soil


Subject to weather

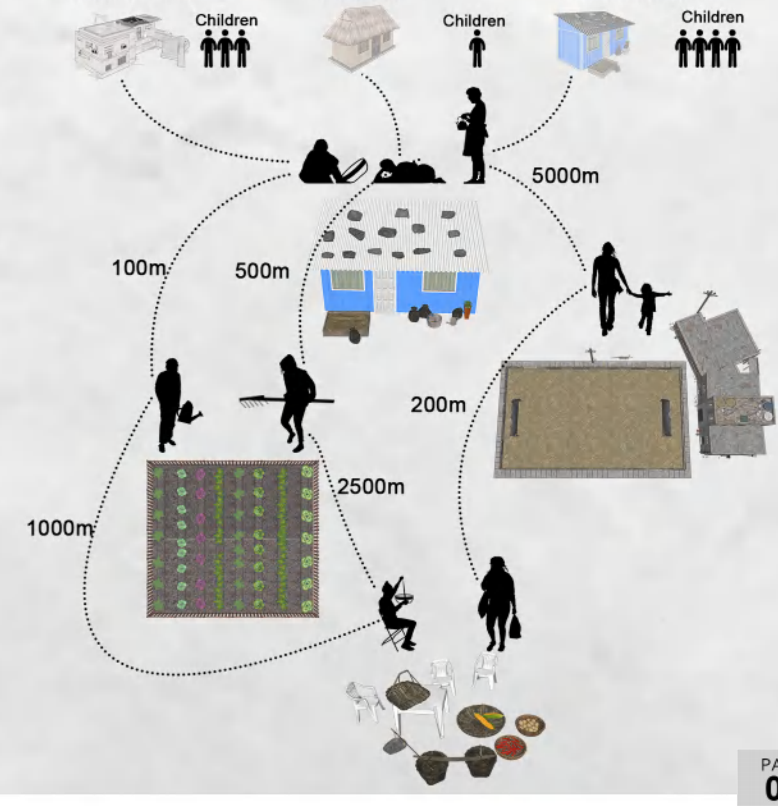

Grain decay

3 RESIDENTS' LIFE


Housewife


Farmer

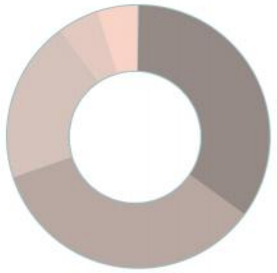

Booth owner



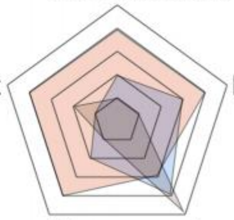
IMPROVING DESIGN OF BUILDINGS

1 BUILDING CHARACTERISTICS

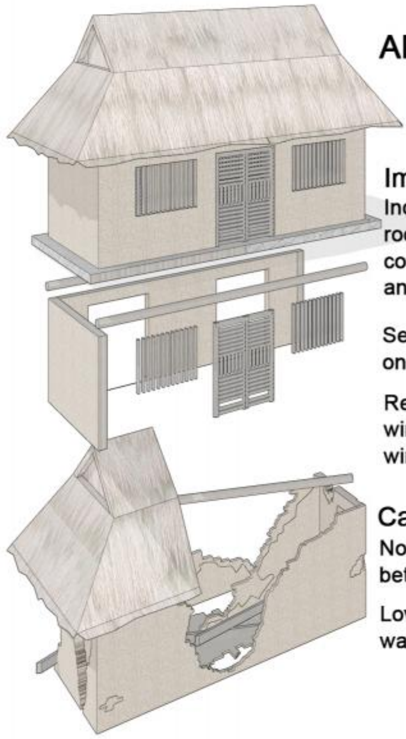
INFLUENCE FACTOR



BUILDING ADAPTABILITY



2 DAMAGE FORMS



ADOBE HOUSE

Improvement

Increase the weight of roof and strengthen the connection between roof and wall

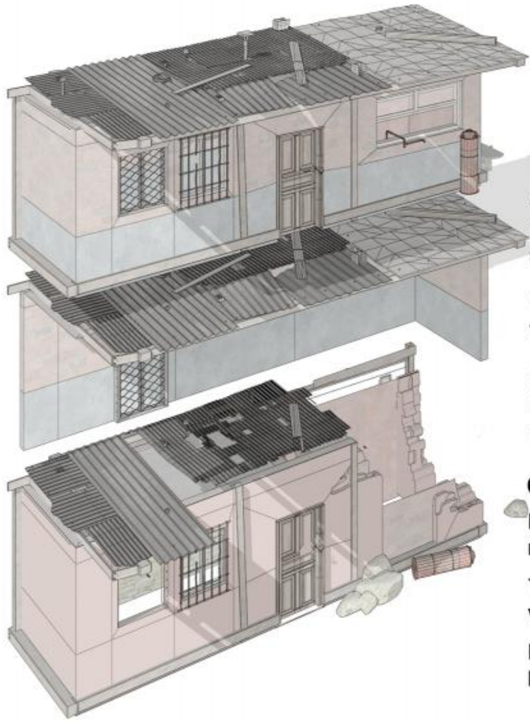
Set damp-proof course on the wall

Reinforce doors and windows and make windproof treatment

Cause

No effective connection between wall and roof

Low strength of bearing wall



BRICK MASONRY HOUSE

Improvement

Solid wall shall be used for key parts and mortar quality shall be guaranteed

Use bolt to strengthen the connection

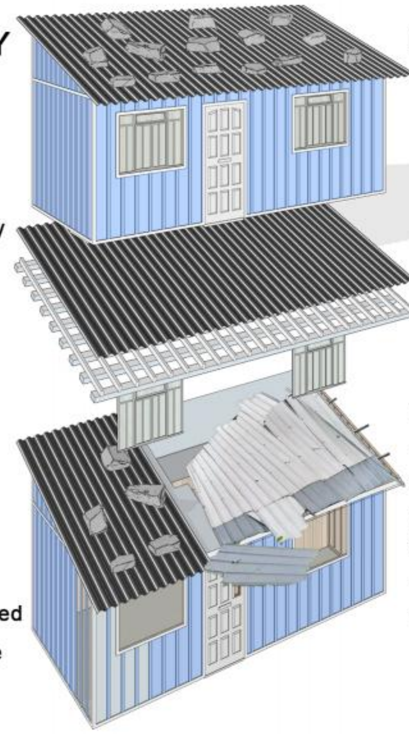
Set bamboo chips, bark and other materials in the wall

Cause

Low strength of masonry mortar

The vertical and horizontal walls are not firmly connected

Poor overall structure of the house



LIGHT STEEL HOUSE

Improvement

Strengthen built connections

Increase roof weight

Reinforce roof construction and increase roughness

Cause

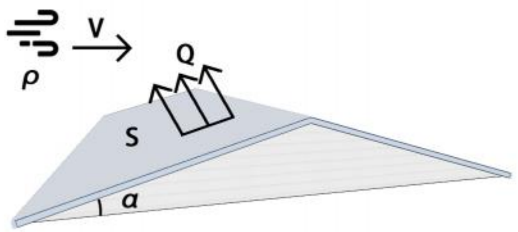
The steel plate on the roof is lifted

Construction distortion, looseness and deformation

The roof is partially bulging or blown away

3 PRINCIPLE OF WIND POWER

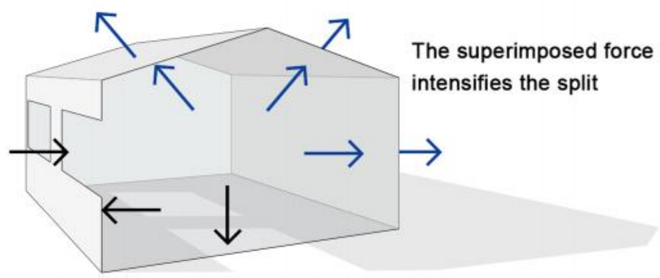
STRESS OF ROOF



$Q=1/2\rho v^2\cos^2\alpha * S$

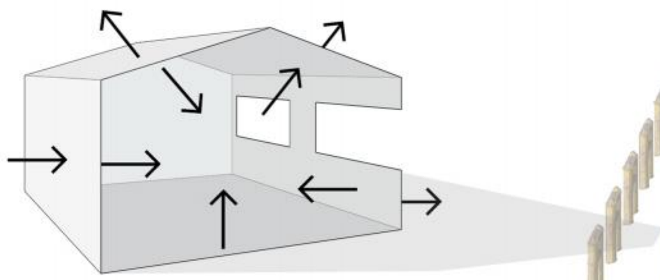
The roof slope angle should be 15 °~25 °

STRESS OF WINDWARD SIDE-OPENING-BUILDING

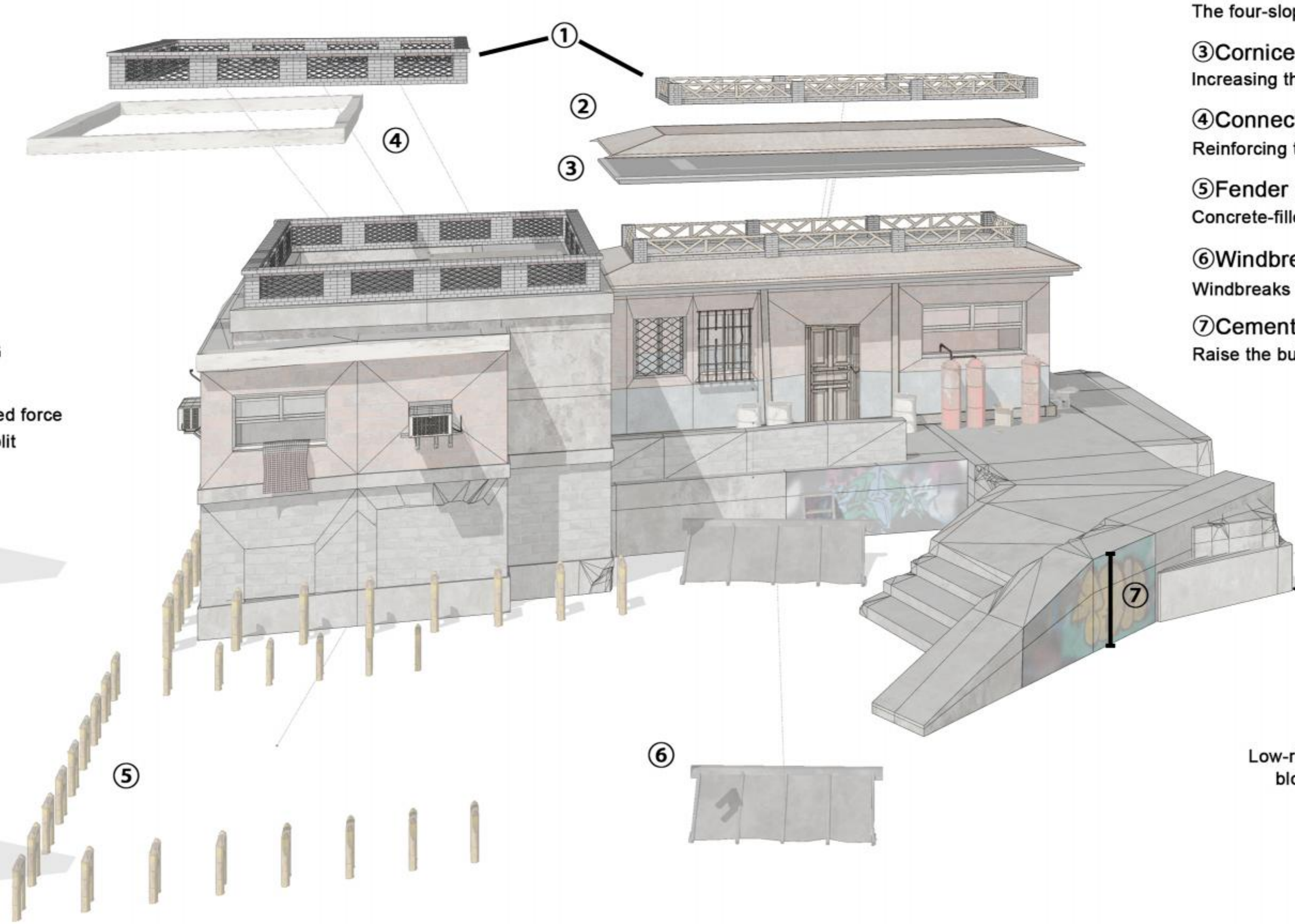


The superimposed force intensifies the split

STRESS OF LEEWARD SIDE-OPENING-BUILDING



4 IMPROVEMENT PLAN



①Parapet
Hollow parapet to reduce the wind suction on the roof

②Four-slope roof
The four-slope roof has better wind resistance than the ordinary herringbone double-slope roof

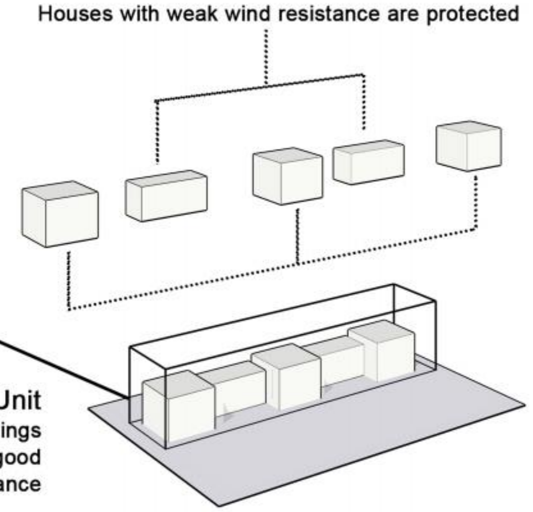
③Cornice
Increasing the cornice length can reduce the suction at the leeward roof cornice

④Connection
Reinforcing the connection between roof and main structure

⑤Fender pile
Concrete-filled steel pipes are used to reduce flood impact

⑥Windbreak
Windbreaks are used to protect plants and property

⑦Cement platform
Raise the building to prevent flood entering

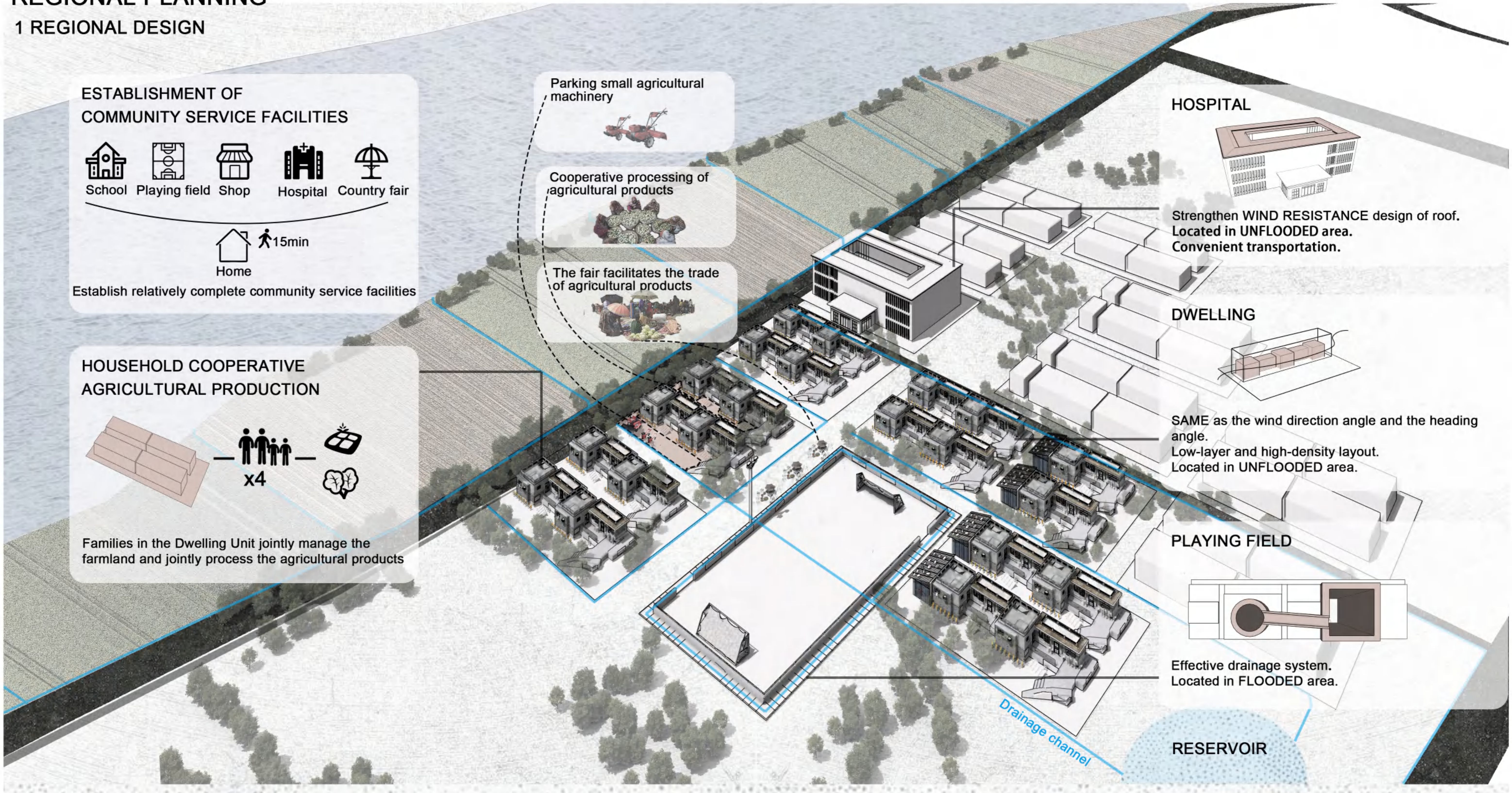


Dwelling Unit

Low-rise and high-density buildings block each other and have good wind resistance

REGIONAL PLANNING

1 REGIONAL DESIGN



ESTABLISHMENT OF COMMUNITY SERVICE FACILITIES

School | Playing field | Shop | Hospital | Country fair

Home 15min

Establish relatively complete community service facilities

HOUSEHOLD COOPERATIVE AGRICULTURAL PRODUCTION

x4

Families in the Dwelling Unit jointly manage the farmland and jointly process the agricultural products

Parking small agricultural machinery

Cooperative processing of agricultural products

The fair facilitates the trade of agricultural products

HOSPITAL

Strengthen WIND RESISTANCE design of roof.
Located in UNFLOODED area.
Convenient transportation.

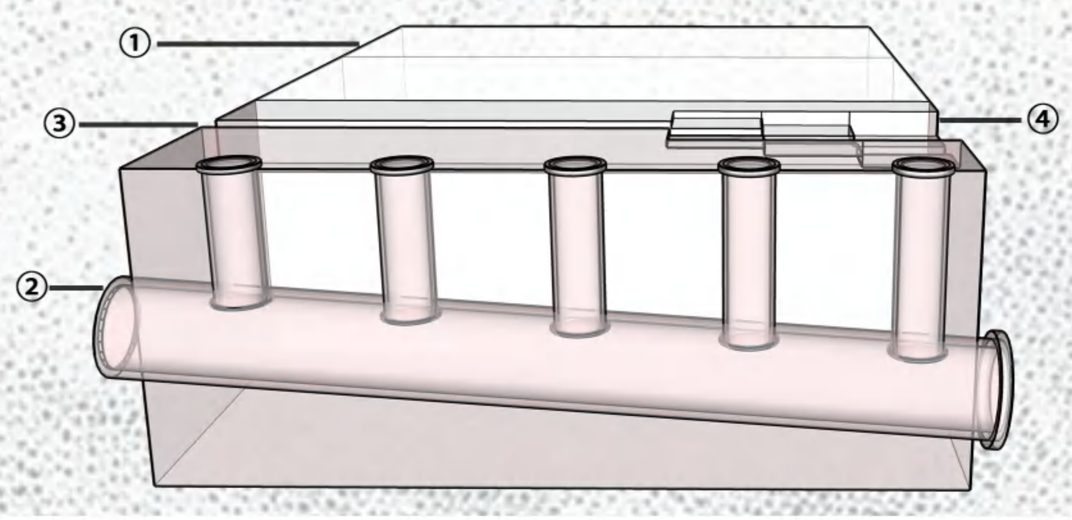
DWELLING

SAME as the wind direction angle and the heading angle.
Low-layer and high-density layout.
Located in UNFLOODED area.

PLAYING FIELD

Effective drainage system.
Located in FLOODED area.

2 DRAINAGE SYSTEM



①Multiple drainage blocks

Flood drainage shall be managed in blocks, and corresponding measures shall be formulated according to the terrain

②Gravity flow drainage system

Use gravity to collect rainwater in towns and discharge it into buzi rivers, and set anti-backflow facilities at the outlet

③Drainage ditch

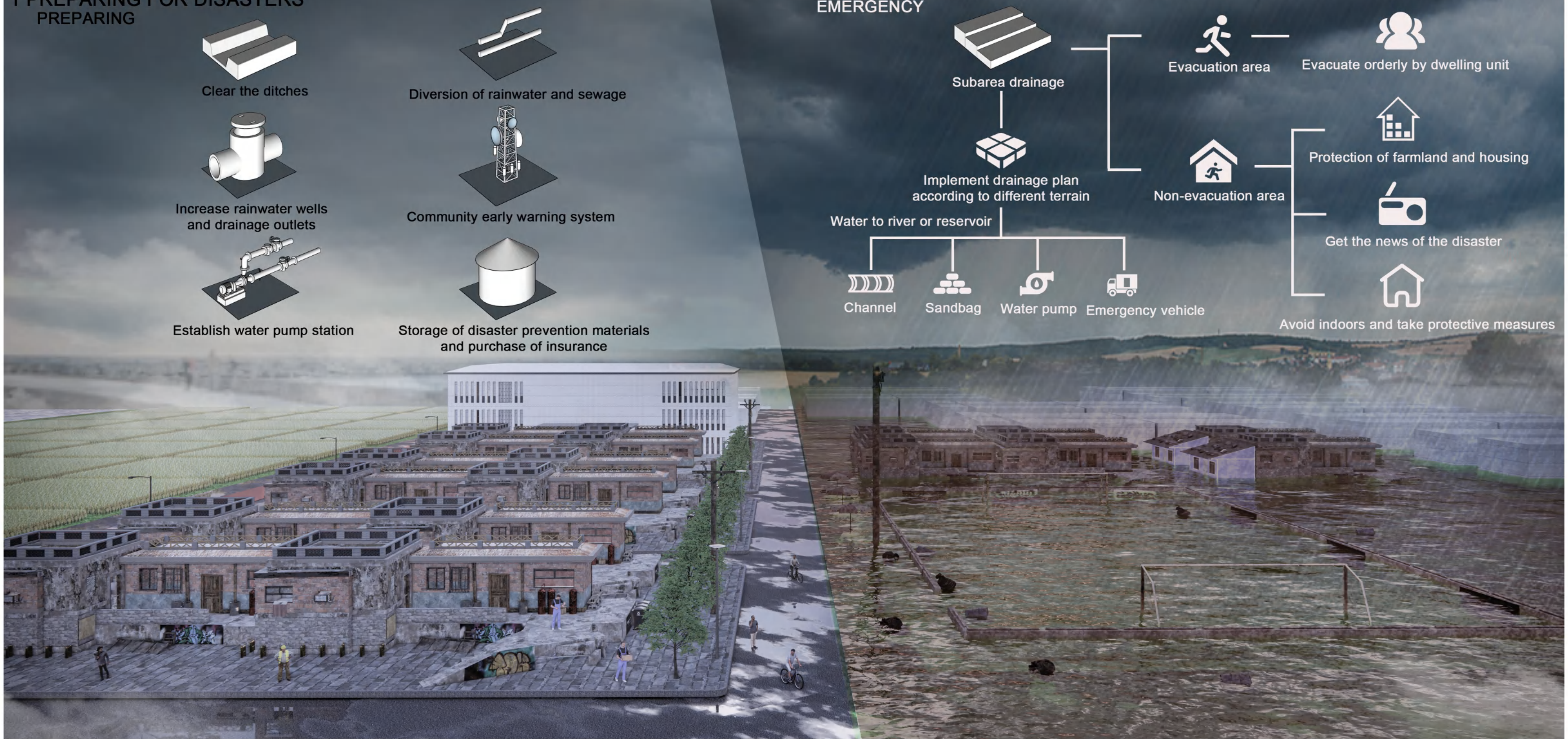
Drainage ditches and other facilities are set beside the road to collect rainwater and collect it into the reservoir

④Stepped reservoir

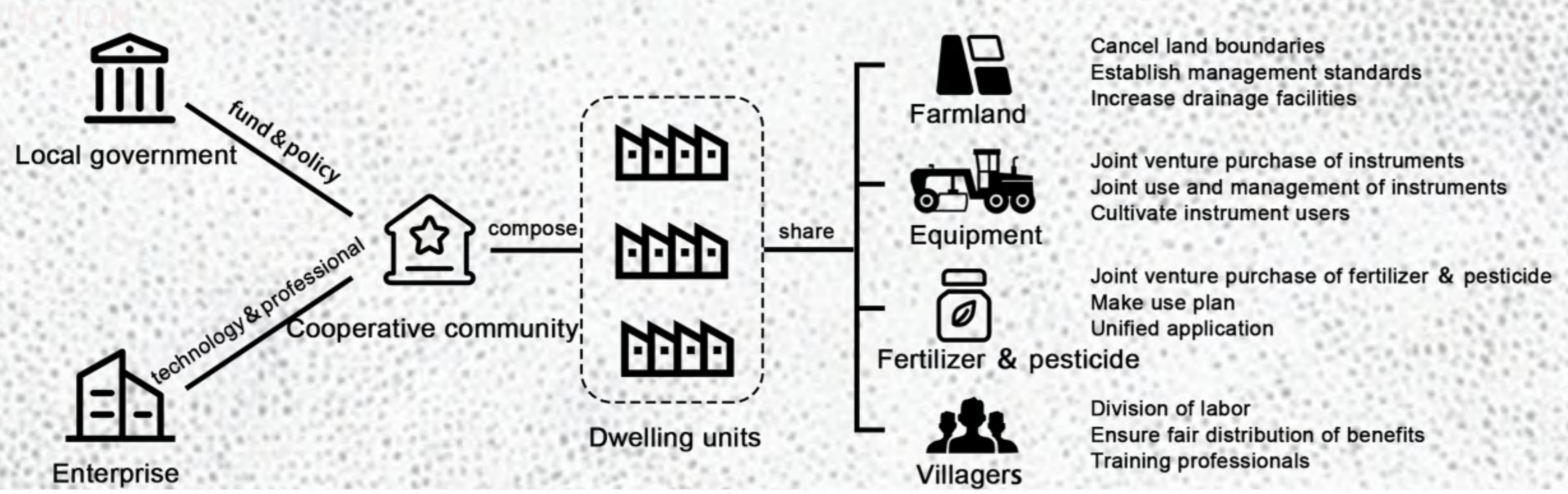
It is used to collect rainwater during drought and increase flood drainage during flood

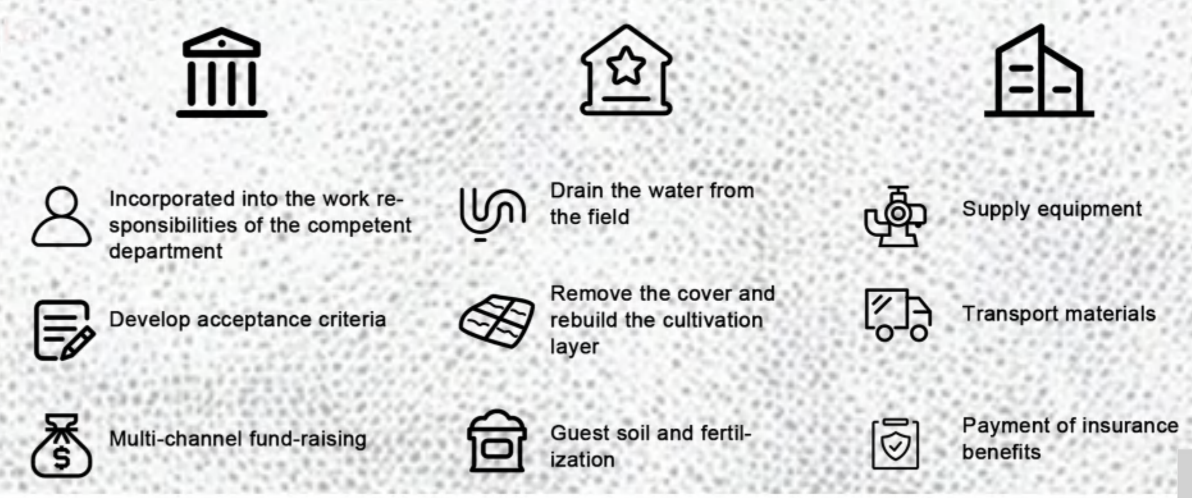
DISASTER PREVENTION DESIGN

1 PREPARING FOR DISASTERS



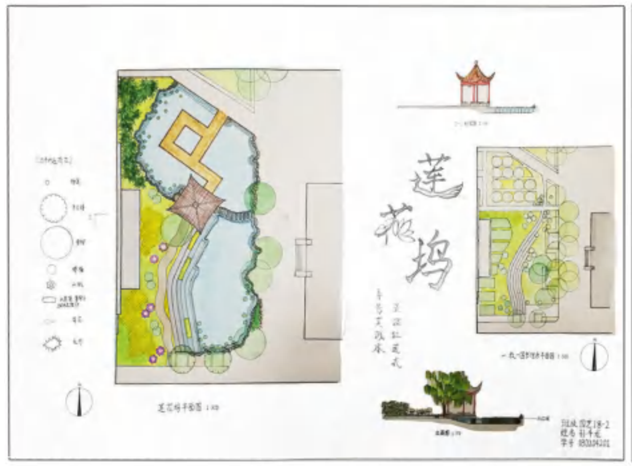
2 AGRICULTURAL PRODUCTION



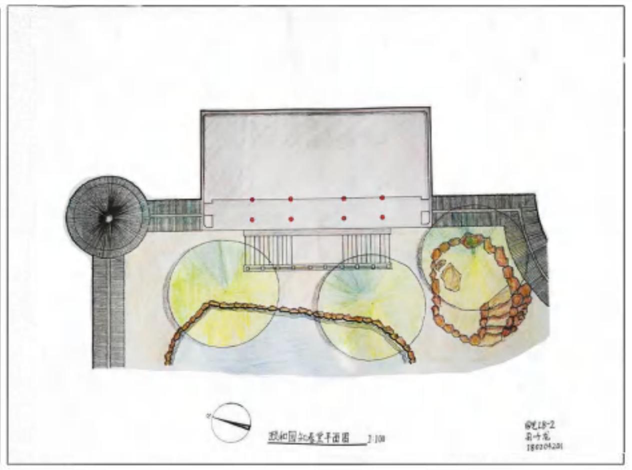


OTHER WORKS

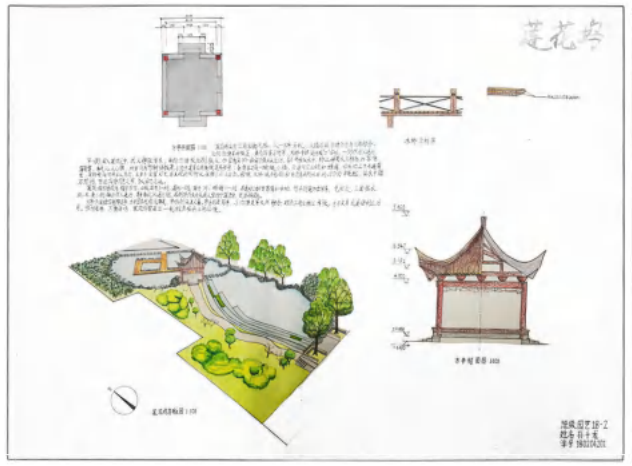
HAND-PAINTED DESIGN



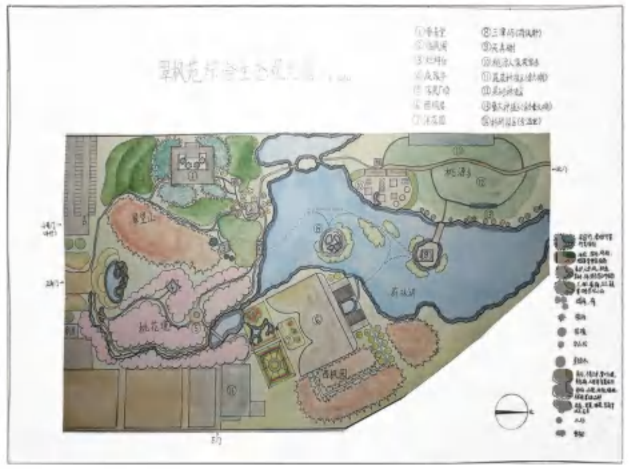
Watercolor Design Drawing(1)



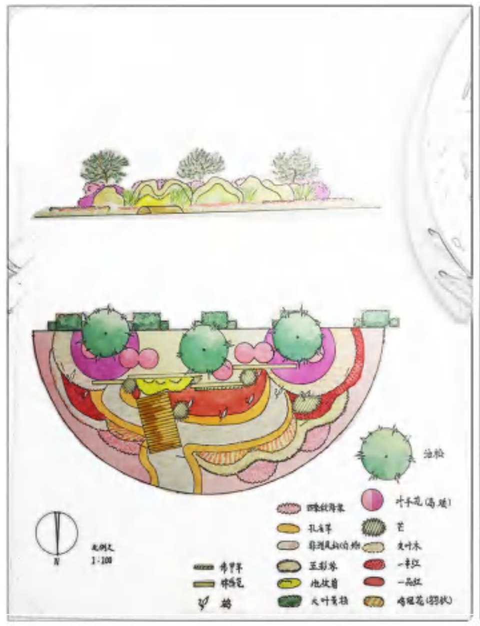
Landscape Survey anc Draw
Apr, 2019



Watercolor Design Drawing(2)
Jun, 2019



Garden Design
Dec, 2019



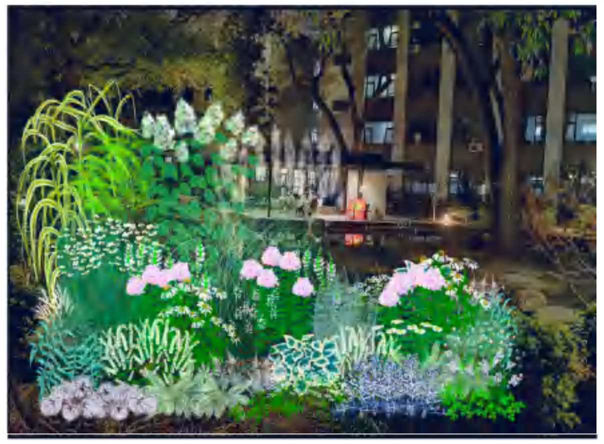
Parterre Design
Oct, 2019

Landscape Survey and Draw
Apr, 2019

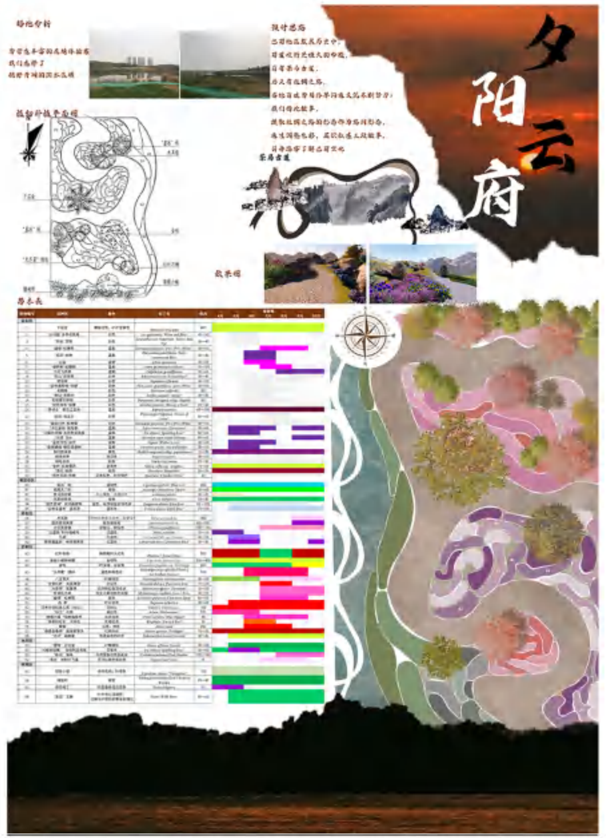
Flower Arrangements
Jun, 2020

Field Practice Oct, 2020
Thanks to my university, teachers, and classmates, I have gained knowledge, skills, friendship, and a love for learning and life from them

FLOWER BORDER DESIGN



Hand Drawn Design of Flower Borders
Apr,2021



Design of Flower Border
May, 2021

LANDSCAPE PRACTICE

Design and Construction of Flower Borders
Apr~Jul, 2021

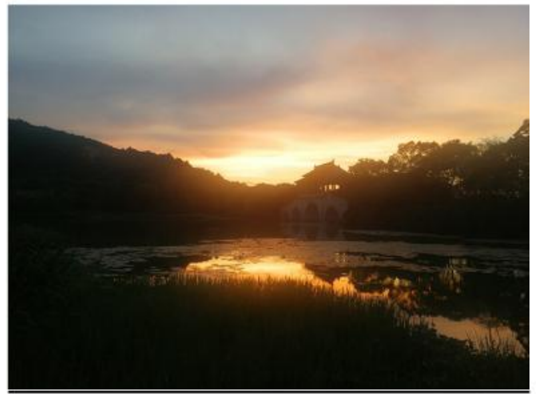
Establishment of Insect Hotels
Aug, 2021~May, 2022

HANDICRAFT

Rock Miniascape
Jun, 2020

PHOTOGRAPHY

Keukenhof Park,Netherlands
Apr, 2017



Wuxi,China
Aug, 2020